



UNIVERSITY OF SRI JAYEWARDENEPURA - FACULTY OF APPLIED SCIENCES
BSc General Degree Second Year First Semester Course Unit Examination – September/October 2024

DEPARTMENT OF PHYSICS

PHY201 2.0 Optics 1

Time: Two (02) hours	No. of questions: 04	No. of pages: 04	Total marks: 100
Answer <u>ALL</u> questions			

1)

- a) Verify the law of reflection using geometric derivation of Huygens principle and optical paths. Note: Explain the geometry where ever it need for the derivation. [10 marks]

- b) The Figure 1A shows a graphical construction made to evaluate the direction of a refracted ray. The line OA represents the direction of the incident ray and the line OR represents the direction of the refracted ray. The line NN' represents the normal to the refracting interface/surface. The symbols γ and β represent the angle of incidence and angle of refraction. The refractive index of the media if the incident ray is n while that of the other media is m . Show that the two rays obey/satisfy the Snell's law (Law of refraction). The two intercept points X and Y gives $\frac{OX}{OY} = \frac{n}{m}$. [05 marks]

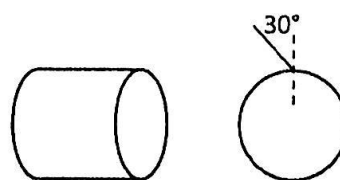
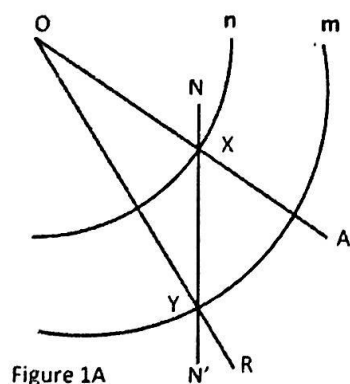


Figure 1B

- c) The incident angle of a light ray, aimed towards the top surface of a cylindrical rod (suspended horizontally) made with glass (refractive index 1.5), is 30° . The plane of incidence is parallel to the circular surface of the rod (See Fig. 1B). And the radius of the rod is 5 cm. Determine the total deviation by a measurement in a graphical construction or using mathematical formula(s) and geometry. [10 marks]

[25 marks]

2) Interferometers are devices that extract information from interference of light waves to measure lengths/distances. There are many types of interferometers and they are used in many different applications. Michelson Interferometry (MI) is one such type of interferometer.

a) **Sketch** a schematic of a MI and label the main items. [05 marks]

b) Using proper diagram(s), **briefly explain** how and under what conditions the circular fringes are generated by the MI. [05 marks]

c) **Briefly explain** what adjustment(s) in the MI is(are) needed to acquire localized fringes and why/how above adjustment(s) change the circular fringes in to localized fringes. [05 marks]

d) **Write** the expression for observing dark and bright fringes in MI, identify the parameters. [05 marks]

e) The Michelson interferometer can be used to measure the index of refraction of a gas by placing an evacuated transparent tube in the light path along one arm of the device. Fringe shifts occur as the gas is slowly added to the tube. Assume 600 nm light is used, the tube is 5.00 cm long, and 160 bright fringes pass on the screen as the pressure of the gas in the tube increases to atmospheric pressure. **Find** the index of refraction of the gas? [05 marks]

[25 marks]

3) Light passing through slit(s) generate light patterns on a screen due to interference. As the number of slits increases some parameters related to the diffraction pattern also changes.

a) **Write** the expression for the intensity of the diffraction pattern,

i. due to Frounhofer diffraction by single slit

[02 marks]

ii. due to double slit, neglecting the effects by individual slits.

[02 marks]

iii. due to double slit, considering the effects by individual slits.

[01 marks]

b) The diffraction grating consists of a large number of equally spaced parallel slits and can be divided in to two types. A transmission grating which have parallel grooves that are transparent to the light and hence act as separate slits. A reflection grating can be made by cutting parallel grooves on the surface of a reflective material. The reflection of light from the spaces between the grooves is specular, and the reflection from the grooves cut into the material is diffuse.

- i. Using proper diagram **illustrate** the order of maximum and zeroth-order maximum and write expression for bright fringes as a function of angle θ identify the parameters [03 marks]
 - ii. Monochromatic light from a helium–neon laser ($\lambda = 632.8 \text{ nm}$) is incident normally on a diffraction grating containing 6000 grooves per centimeter. **Show** that only three orders of maximum can be observed in this setup. [10 marks]
- c) The ability of optical systems to distinguish between closely spaced objects is limited because of the wave nature of light. Considered two noncoherent monochromatic sources S_1 and S_2 of wavelength λ at a distance very far from a narrow slit of width a .
- i. Write the Rayleigh's criterion for resolution [01 marks]
 - ii. Write an expression for the limiting angle of resolution for a circular aperture [01 marks]
 - iii. In Sri Lanka standard analog television picture (non- HDTV), also known as PAL, is composed of approximately 540 visible horizontal lines of varying light intensity. Assume your ability to resolve the lines is limited only by the Rayleigh criterion, the pupils of your eyes are 5.00 mm in diameter, and the average wavelength of the light coming from the screen is 550 nm. Calculate the ratio of the minimum viewing distance to the vertical dimension of the picture such that you will not be able to resolve the lines. [05 marks]
- [25 marks]
- 4) Light can be polarized using both reflection and refraction. Malus law and Brewster's law are two laws that can be used to solve problems related to polarization of light.
- a) **Sketch the graph** of intensity against the angle of polarizer, for following conditions, with the line style in brackets, on the same figure. [05 marks]
 - i. Source is unpolarized light source (solid; ———)
 - ii. Source is polarized light source (dash; - - - - -)
 - b) **Sketch the graph** of intensity of the monochromatic reflected light against the angle of the incidence for the following conditions, with the line style in brackets on the same figure. [10 marks]
 - i. Observe without a polarizer (solid line; ———)
 - ii. Observe with an ideal polarizer oriented parallel to the interface (dashed line; - - - - -)
 - iii. Observe with an ideal polarizer oriented perpendicular to the interface (crossed line; + + + + +)

- c) Assume you have a vertically polarized light source. You need to rotate the plane of polarization of this polarized light beam by a total of 45.0° . You use a sequence of nine (09) ideal polarizing filters, each with its axis making the same angle with the axis of the previous filter. Show that you can get more than 85% higher intensity compared to using a single ideal polarizer at 45.0° by using above method.

[10 marks]

[25 marks]

***** End of the Question Paper *****

Few equations/formulas that may be useful (symbols represents their usual parameters)

$$y = 2a \cos \frac{k\Delta}{2} \sin \left[\omega t - k \left(x + \frac{\Delta}{2} \right) \right]$$

$$I \approx b^2 I^2 \frac{\sin^2 \beta}{\beta^2} \frac{\sin^2 \gamma}{\gamma^2}$$

$$\beta = \frac{\pi b(\sin i + \sin \theta)}{\lambda}$$

$$I \approx A^2 = 2a^2(1 + \cos \delta) = 4a^2 \cos^2 \frac{\delta}{2}$$

$$\frac{n}{s} + \frac{n'}{s'} = \frac{n' - n}{r}$$

$$A = atrt' \frac{1}{1 - r^2}$$

$$\sin A + \sin B = 2 \sin \frac{1}{2}(A + B) \cos \frac{1}{2}(A - B)$$

$$\frac{n}{s} + \frac{n'}{s_h} = \frac{n' - n}{r} + \left[\frac{h^2 n^2 r}{2f'n'} \left(\frac{1}{s} + \frac{1}{r} \right)^2 \left(\frac{1}{r} + \frac{n' - n}{ns} \right) \right]$$

$$\tan \theta = \frac{a_1 \sin \alpha_1 + a_2 \sin \alpha_2}{a_1 \cos \alpha_1 + a_2 \cos \alpha_2}$$

$$I = I_{max} \left(\frac{\sin \left(\frac{\pi a \sin \theta}{\lambda} \right)}{\frac{\pi a \sin \theta}{\lambda}} \right)^2$$

$$I = I_{max} \cos^2 \left(\frac{\pi d \sin \theta}{\lambda} \right)$$